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Editorial Office
BMC Medical Research Methodology

Dear Editors,

I am submitting the manuscript “Ecological bias distorts effect sizes in federated COVID-19 analysis: evidence from 42 million patient records” for consideration as a Research Article in *BMC Medical Research Methodology*.

This paper quantifies how ecological bias distorts effect sizes when multi-site studies pool site-level summaries instead of individual patient data. Using 42 million COVID-19 records from the United States (CDC) and Mexico, we show that ecological regression produces linear slopes incommensurable with the individual-level odds ratios they purport to estimate—a different quantity, not a noisy estimate. Applied to the 4CE consortium’s neurological dataset (21 hospitals, 6 countries), meta-analytic methods diverge by 350-fold, and a multivariate consistency test detects coordinated heterogeneity invisible to standard approaches.

The paper contributes to the methodological literature in three ways: (1) it provides large-scale empirical quantification of ecological bias in federated clinical data, a setting where the fallacy is routinely acknowledged but rarely measured; (2) it introduces structural diagnostics (a multivariate consistency test with a formal degeneracy proof) for detecting when site-level pooling is unreliable; and (3) it includes ten pre-registered supplementary analyses with explicit falsification criteria, two of which fired, demonstrating transparent reporting of both confirmatory and disconfirmatory results.

The datasets used are publicly available (CDC Case Surveillance, Mexico Datos Abiertos), all code is available on GitHub, and the analysis protocol was frozen before results were examined. The manuscript has not been submitted elsewhere and is not under consideration at another journal.

I believe this work is well suited to BMC Medical Research Methodology’s focus on statistical methods for health research, and I hope the editors and reviewers will find it a useful contribution.

Sincerely,

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